

Metal Casting Processes and Their Applications

Introduction to Foundry Processes

We can shape metal into useful products by several methods; and casting offers the minimum distance between the raw material and the finished product. Casting processes basically consist of pouring a molten metal in a mould cavity where upon solidification the metals assumes the shape of the mould cavity. This product is called a casting.

Sand casting

The important terms used in metal casting are briefly explained below;

1. **Pouring basin:** A small funnel shaped cavity at the top of the mold into which the molten metal is poured.
2. **Sprue:** The passage through which the molten metal, from the pouring basin, reaches the mold cavity. In many cases it controls the flow of metal into the mold.
3. **Gate:** A channel through which the molten metal enters the mold cavity.
4. **Riser:** A column of molten metal placed in the mold to feed the castings as it shrinks and solidification. This is also known as "feeder head".
5. **Runner:** The channel through which the molten metal is carried from the sprue to the gate.
6. **Pattern:** It is the replica of the final object to be made. The mold cavity is made with the help of a pattern
7. **Flask:** A metal or wood frame, without fixed top or bottom, in which the mold is formed.
8. **Parting Line:** This is the dividing line between the two molding flasks that makes up the mold. It divides the mould into two halves or it is so chosen that more volume of casting goes to the drag.
9. **Moulding Sand:** Sand, which binds strongly without losing its permeability to air or gases. It is a mixture of silica sand, clay and moisture in appropriate proportions.
10. **Core:** A separate part of the mold made of sand mixed with organic binders and generally baked to develop strength. These core are used to create holes, openings and various shaped cavities in the castings

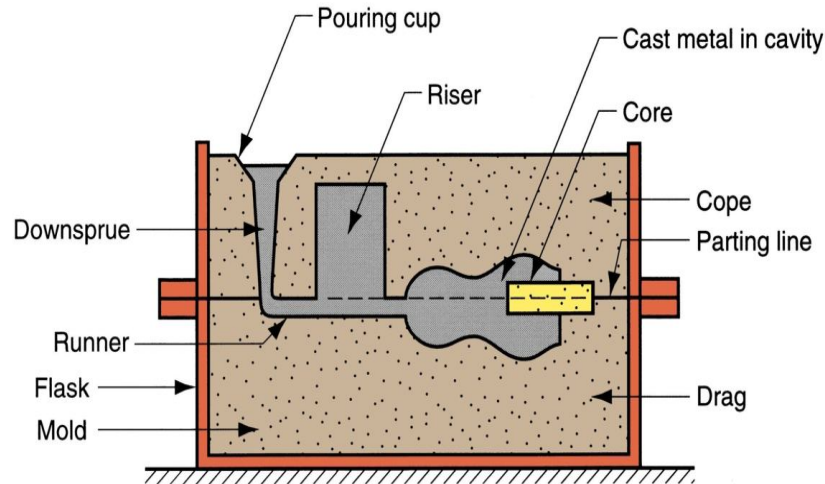


Figure-Cross-sectional view of sand mold

Steps in Making Sand Castings

There are six basic steps in making sand castings:

1. Patternmaking
2. Core making
3. Molding
4. Melting and pouring
5. Cleaning

Pattern and allowance

Pattern Material

- 1. Wood:** Wood is the most widely used material for pattern making because it has following the properties,
 1. It is easily available
 2. It has low weight.
 3. It is easy to work with wood.
 4. It is relatively cheap.
- 2. Metals and alloys:** Metals used for pattern making are cast iron, aluminum and copper alloy. Choice of metals depends widely upon quality and quantity of mould to be

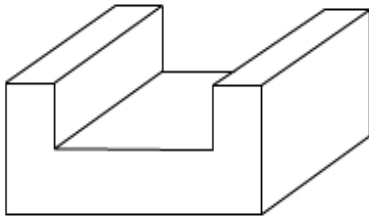
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produced. For large scale production cast iron is used as it is cheap, easy to machine and wear resistant. Aluminum alloys are used for small scale production as it has lesser wear resistance.

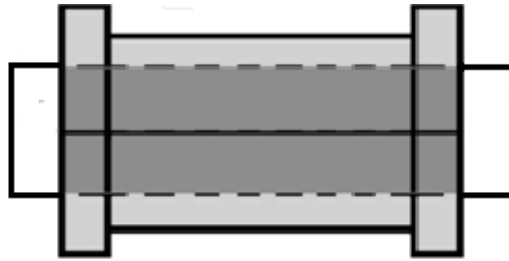
Types of Pattern

1. **Single Piece Pattern or solid pattern:** solid pattern is made in single piece and is used when shape is relatively simple and does not create withdrawal problem, it is inexpensive and made from wood
2. **Split Pattern or Two Piece Pattern:** Some pattern cannot be made in single piece due to difficulty in removing pattern from mold. Pattern is split into two parts by parting lines cope and drag. Tapered pins in the cope with holes in drag for precise positioning.
3. **Gated Pattern:** This is an improvement over the simple pattern where the gating and runner system is integral with the pattern. This would eliminate the hand cutting of the runners and gates and help in improving the productivity of a molder
4. **Loose Piece Pattern:** Loose piece patterns are used when some irregularity is present on the contour of the pattern, making a difficulty in withdrawing the pattern from the mould. So to overcome this difficulty the irregular part of the pattern is held loose against the contour of the pattern using dowel pins or wire. By using such arrangement during the withdrawal we can first withdraw the main pattern which is regular in shape and then the loose pieces can be removed without damaging the contour of the casting from the gap provided by the main pattern.
5. **Match plate:** Provide a strong and a rigid mounting for pattern and help in eliminating the mismatch between the cope and drag half of the mould during the assembly of machine moulds. Usually all high production work on moulding machines uses the match plate pattern.
6. **Follow board pattern:** Used with single or multiple gated pattern. Pattern requiring follow boards are difficult to make as split pattern. The pattern rests in the follow board up to the parting line. In this position, it serves as a moulding board for the first moulding operation.
7. **Sweep Pattern:** many moulds of regular cross-section are made by the use of sweep patterns as shown in figure. It eliminates expensive pattern construction.

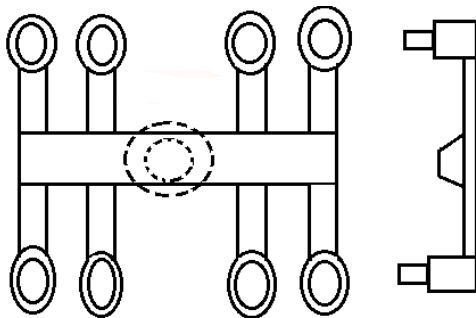
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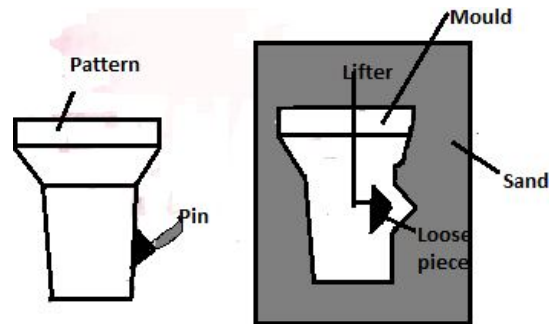
(a) Solid



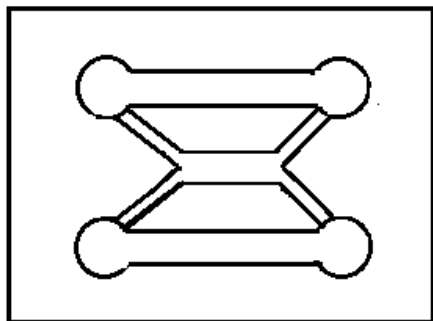
(b) Split



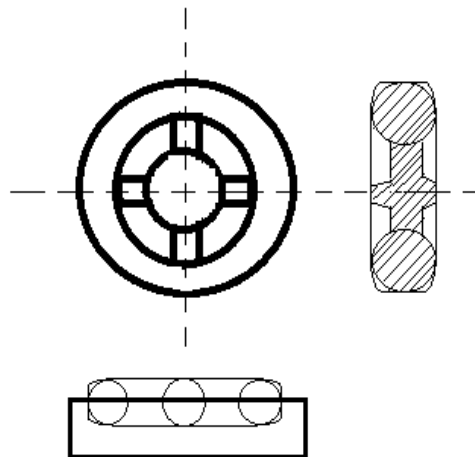
(c) Gated



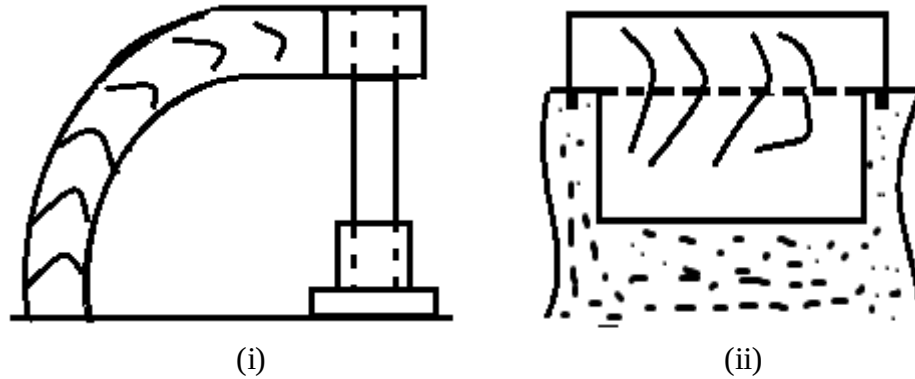
(d) loose piece



(e) Match plate



(f) Follow board



(g) Sweep patterns

Figure-Types of patterns

Pattern allowances

Dimensions of pattern differ from the final casting. This difference of dimension is called allowances and is provided to ensure easy withdrawal of pattern and get dimensionally correct finished casting. There are mainly five types of allowances provided on a pattern.

1. **Shrinkage allowance:** when metal cools from pouring temperature to room temperature, it contract. This contraction takes place in three stages-
 - a. Contraction of liquid from pouring temperature to freezing temperature (liquid contraction).
 - b. Contraction due to Change of phase from liquid to solid known as solidification contraction (solid contraction)
 - c. Contraction of casting from freezing temperature to room temperature known as solid contraction (solid to solid contraction)

First two contractions are compensated by riser and gates while last contraction is considered for shrinkage allowance. Stages of shrinkage are shown in given figure.

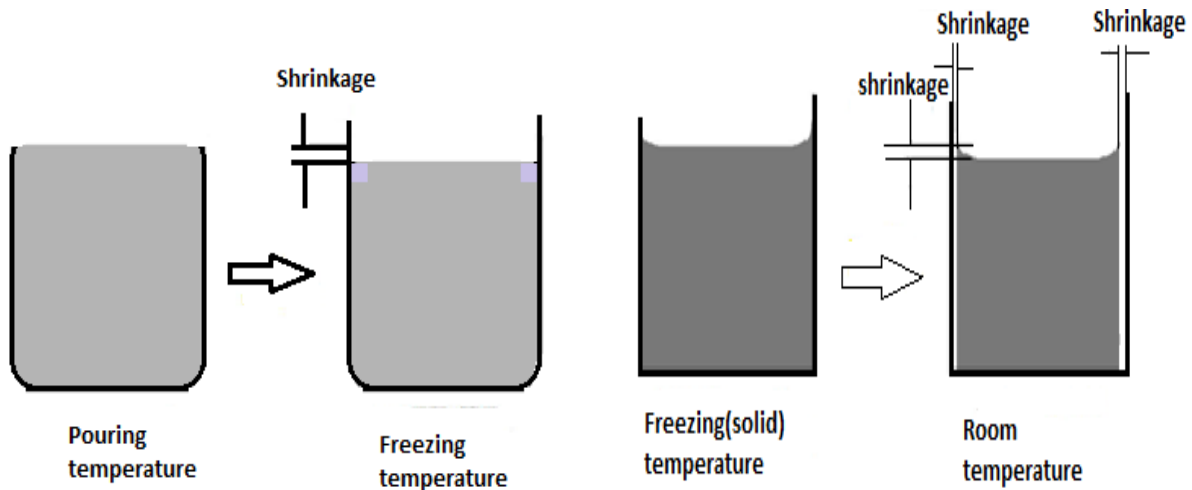


Figure- Stages of shrinkage

2. **Draft Allowance:** The draft allowance is defined as the tapering which is provided on the pattern. When the sides of the pattern are deeper, it may damage the walls of the pattern while with drawing it from the mould cavity. To overcome this problem a taper or slight inclined surface is furnished on the sides of the pattern for the easy withdrawal of pattern from the mould. Table below shows the values of taper to be provided for easy removal.

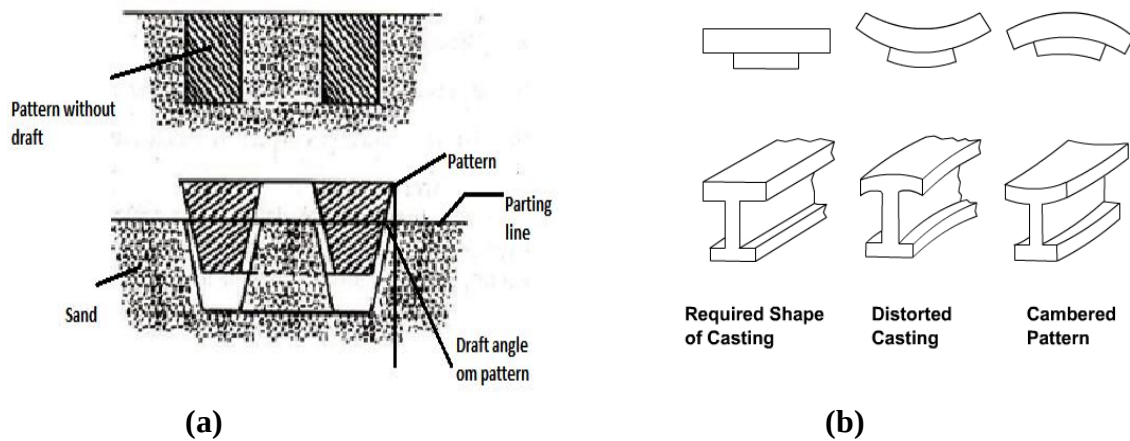


Figure (a) Draft and (b) Distortion allowance

3. **Distortion or camber allowance:** This is applicable to casting of irregular shape such as “V”, “U” and “T” section. When such castings are allowed to cool they are distorted due to non-uniform contraction. To eliminate this, the shape of pattern should be given allowance of equal amount in opposite direction.
4. **Machining allowance:** when an object is prepared using casting, the surface finish we get is very bad and rough. To overcome this issue some extra metal is provided on the surface to machine and make the surface smooth. This allowance is known as machining allowance.

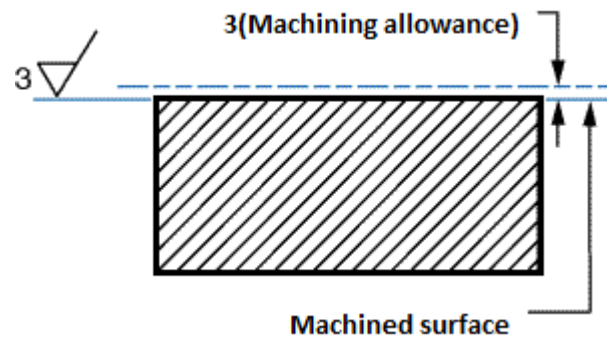


Figure -Machining Allowance

5. **Shake or Rapping allowance;** When a pattern is to be drawn from the mould, it is first rapped or shaken, by striking over it from side to side, so that its surface may be free of the adjoining sand wall of the mould. As a result of this the size of the mould cavity increases a little and a negative allowance is to be provided in the pattern to compensate the same. However, it may be considered negligible for all practical purposes in small and medium sized casting.

Desirable properties of moulding sand

1. Inexpensive in bulk quantity
2. Retain properties through transportation and storage
3. Uniformly fills a flask or container
4. Compacted or set by simple methods
5. Sufficient elasticity to remain undamaged during pattern withdrawal
6. Withstand high temperatures and maintain its dimensions until metal solidifies Sufficient permeable to allow the escape of gases
7. Sufficiently dense to prevent metal penetration
8. Sufficiently cohesive to prevent washout of mold

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9. Chemically inert to metal being cast
10. Good collapsibility to permit easy removal and separation of casting
11. Can be recycled

Constituent of moulding sand

12. Silica (SiO_2) = 86%-90%
13. Aluminum (Al_2O_3) = 4%-8%
14. Iron oxide (Fe_2O_3) = 2%-5%
15. Clay as binder
16. Moisture to provide elasticity

Gating system

Gating system shown in figure provides a passage for the molten metal to enter the mould cavity, fill it completely and rise in the riser. The system is composed of:

- | | |
|----------------|----------|
| 1. Pouring cup | 4. sprue |
| 2. Runner | 5. gates |
| 3. Riser | |

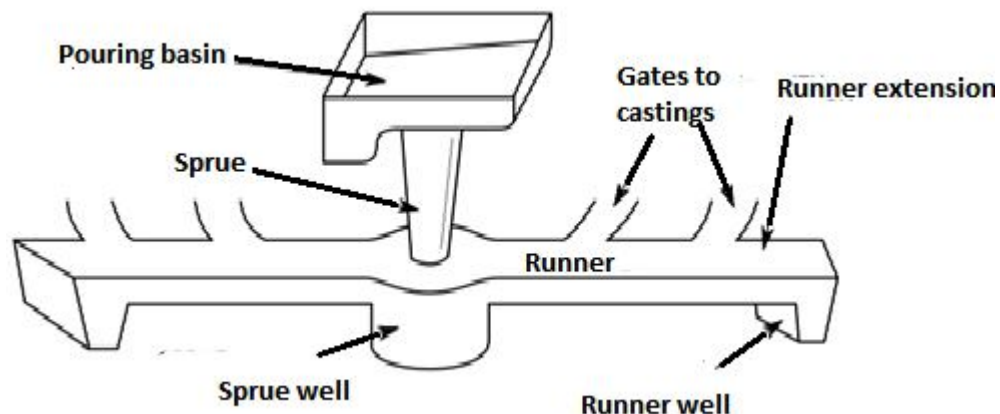


Figure -Component of gating system

Basic Requirement of a gating system

1. A gating system should avoid sudden or right angle changes in direction as it causes:
 - Mould erosion
 - Turbulence
 - Gas pick-up
2. Fill the mold cavity completely before freezing

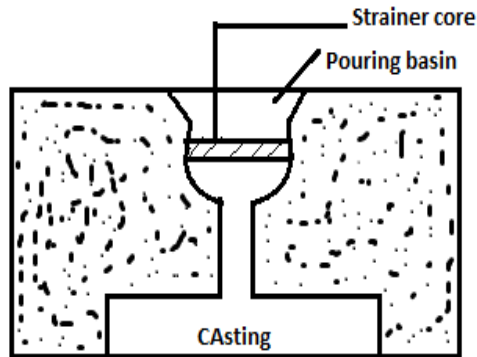
3. Help to promote temperature gradients favorable for proper directional solidification
4. Should incorporate traps for separation of non-metallic inclusions(slag, broken sand, etc)
5. Should regulate proper rate at which metal enters into the mould cavity.
6. Be practical and economical to make.
7. The system should consume least amount of metal

Type of Gates

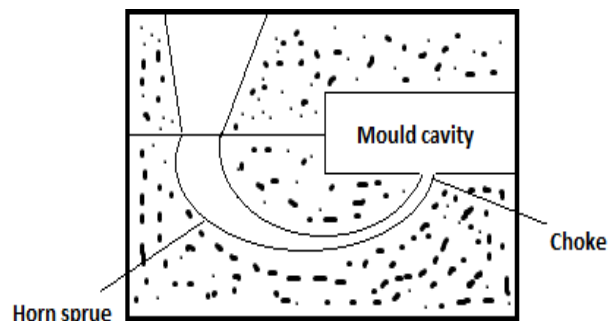
1. **Top gating system:** - In this type of gate the entire molten metal enters the mould cavity at the top, with the result the hottest metal remains at the top of the casting and therefore proper temperature gradient is formed to enable directional solidification of casting from the bottom towards the riser. The gate itself serves as the riser.

Drawback:-

1. Erosion of mould takes place.
2. Turbulence occur in gate
3. Excessive absorption of gases takes place
4. Not favored for non-ferrous castings



(a)



(b)

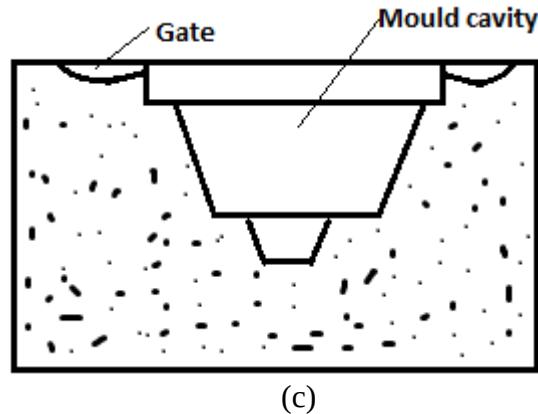


Figure (a) Top gating (b) Bottom gating and (c) Parting line gating system

2. **Bottom gating system:** - In this type of gate the molten metal enters the mould cavity at or near its bottom. It provides a smooth flow of molten metal reduces considerably the erosion of mould and core and minimize turbulence. Inside the cavity the metal rises gently upwards.

Drawback:-

- a. In large and deep casting, the metal cools gradually as it rises up. With the result an unfavorable temperature gradient is produced i.e. the hottest metal remaining at the bottom and coolest metal at the top.
- b. The directional solidification adversely affected.

3. **Parting line gating system:** - These gates are the simplest both in nature and construction. In this type of a gate the metal enters the mould cavity at the parting line. Such gates are very commonly employed and are found to give very satisfactory service except when the mould is very deep.

Drawback

1. It is not suitable for deep mould as the metal has to fall on the mould bottom through a considerable distance between the gate and the bottom.
2. Dropping of metal from such height cause erosion or washing of mould.
3. In non-ferrous metal this drop promotes dross and air absorption by metal, leading to an inferior casting.

Classification of Casting Defects

A large number of defects occur in sand castings produced through various methods. The major factor responsible for these defects includes:

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1. Design of casting
2. Pattern design
3. Mould and core materials used
4. Mould and core system design
5. Gating and feeder head design
6. Melting and pouring
7. Cast metal metallurgical factors

Some of the most common defects are discussed below with their causes and appearances. The defects are discussed in the order of their importance.

Surface Defects:

1. **Wash;** It is low projection on the drag face of a casting that occurs due to sand being washed away due to improper molten metal flow during pouring. The sand that is washed away appears at some other location as sand inclusion. Figure (a)
2. **Rat tails or buckles:** Slight compression failure of a thin layer of moulding sand causes rat tails, a more severe compression failure causes buckles (due to buckling of sand). Figure (b).
3. **Swell:** Swell occurs due to enlargement of mould cavity due to ferrostatic pressure. Figure (c)
4. **Misrun:** This defect occurs due to incomplete mould filling due to low pouring temperature, inadequate metal supply, and improper gating design. This defect determines the minimum thickness that can be cast for a given metal, superheat and type of mould. Figure (d)

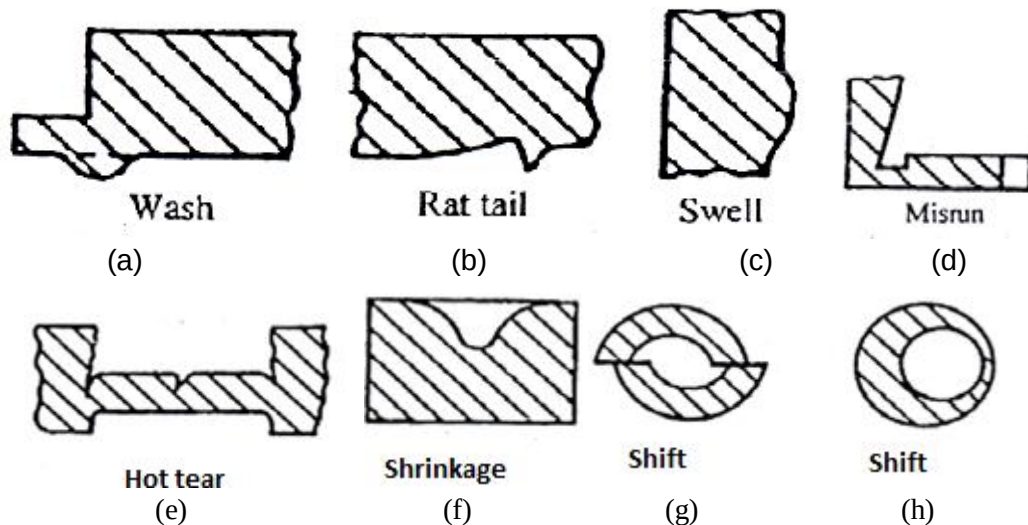


Figure -various casting defects appear on surface

5. **Hot tear and shrinkage:** Hot tears are ragged edge cracks that occur due to tensile stresses during solidification. They occur due to excessive mould hardness

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discontinuity in the metal casting due to hindered contraction and occur just after the metal has solidified. It is shown in figure (e) and (f). Improper metallurgical and pouring temperature control causes this defect.

6. **Shift/Mismatch:** It is due to shift of individual parts of casting due to wrong assembly of cope and drag of mould cavity, with respect to each other figure (g) and (h)

Internal Defects Blow holes, Porosity, Pin holes, Inclusions, Dross etc

1. **Blow Holes:** Blow holes appear as cavities in a casting. They may appear at the cast surface as open blows and when they occur below the surface they are called simply as blows. They occur due to entrapped gases in the metal when vent holes are not properly provided or sand porosity is poor. Figure (a).

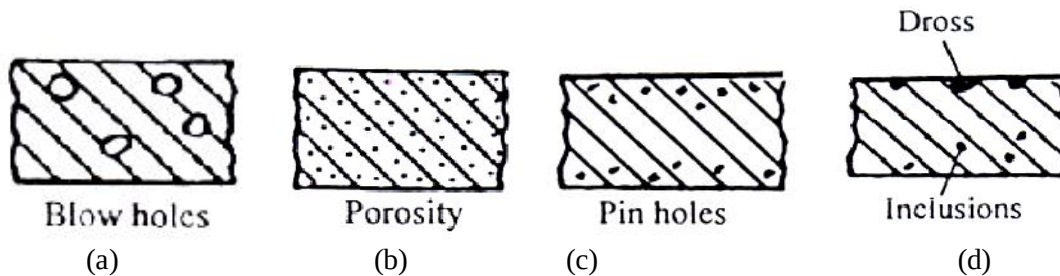


Figure -various casting defects (internal)

2. **Porosity:** This porosity occur due to gases mainly hydrogen absorbed by molten metal which are released on cooling. Effect of micro porosity on metals ductility is shown in figure (b). Porosity may also caused by shrinkage.
3. **Cold shut:** A cold is an interface within casting that formed when two metal streams meet without complete fusion. This defect occurs when the metal streams meet at temperatures low enough to cause fusion

Die Casting:

Die casting is a precision casting technique in which liquid metal is injected into metallic die under high pressure. It is used to for production of fine surface, complex shapes and narrow sections. Die casting is done by

1. **Hot chamber die casting:** There is a pump consisting of plunger and cylinder both made of refractory material. The liquid metal fills the refractory cylinder through the hole. When plunger moves down the metal in the cylinder is forced into the mould cavity through the goose neck. For zinc castings the pumping pressure is up to 40 MPa.

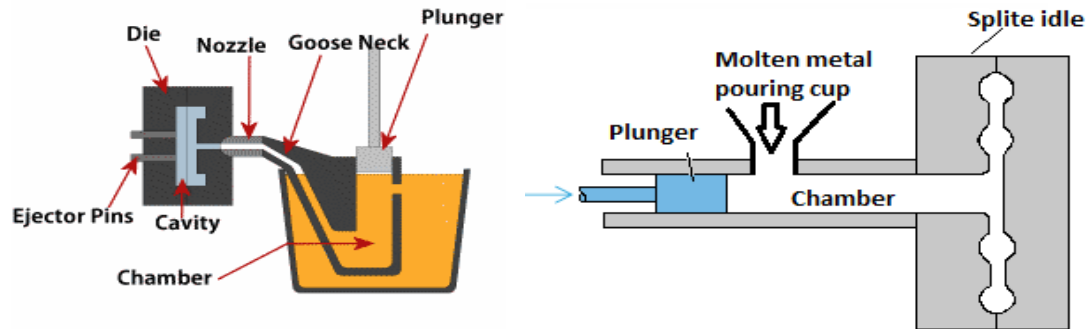


Figure (a) Hot Chamber dies casting (b) Cold chamber die casting

2. **Cold chamber die casting:** A measured quantity of liquid metal is poured in refractory cylinder. Plunger moves and pushes this liquid metal into mould cavity. Pressure exceeding 40 MPa are used for magnesium and aluminum alloys

Cupola Furnace

The majority of gray and white cast iron are melted in cupola furnace. As shown in figure 4.11, the cupola essentially is a vertical cylindrical steel shell with a refractory lining, supported on heavy steel or iron legs so that its lower end 3 to 5 feet above the floor. The lower end is closed by hinged doors that can be opened to permit cleaning the inside. A sand bottom, 4 to 6 inches deep and sloping downward to the pouring spout, is prepared on top of the closed bottom doors. The pouring spout has taphole 1 to 2 inches in diameter.

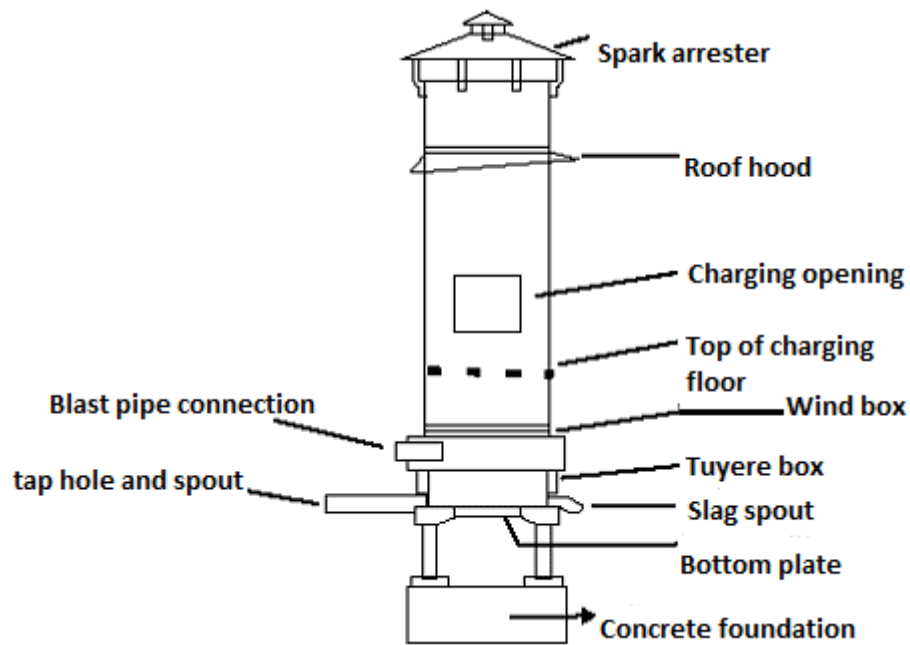


Figure-Cupola furnace

After the sand bottom has been prepared, kindling wood is placed on the bottom and a bed of coke on top of this. The wood is ignited and the coke bed is increased until it is 18 to 48 inches in depth, depending on the size of the cupola. Natural draft is supplied through the tuyeres, which are openings around the periphery of the cupola about 15 to 20 inches above the top of the sand bottom. When the coke bed has been built up to the desired depth and is thoroughly ignited, alternate charges of coke and iron are added through the charging door located about half the distance up the height of the cupola. The metal charge may be either pig iron or scrap. About one part of coke, by weight, is used to eight or ten parts of iron. In addition, small amounts of limestone, of other materials; may be added to act as a flux and to increase the fluidity of the metal. It is possible to add certain alloy-rich materials to alter the chemical composition of the final product, although such additions often are added in the ladle at the time the cupola is tapped.

After the iron and coke charges have been built up to the level of the charging door, the charge is heated for about half an hour under natural draft. The natural draft openings then are closed, and the lower is started. Under forced draft, the temperature in the cupola rise rapidly, and the iron charge start melt and run down through the coke. Final melting occurs in the zone just above the tuyeres. Molten metal should start to appear within ten minutes after the air blast is turn on. Quartz- covered peepholes make it possible to see the metal flowing down through the incandescent coke. As soon as the first metal is starting to flow, the taphole is plugged with a conical- shaped plug of fire clay.

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The molten metal collects in the lower portion of the cupola between the sand bottom and the slag spout, located slightly below the tuyeres. Excess slag floats on top of the molten metal and can be drained off through the slag hole to obtain cleaner metal. When a sufficient amount of molten metal has collected, the air blast is turned off temporarily, the clay plug is punched out, and the metal is permitted to flow out into a pouring or holding ladle. When sufficient metal has been removed, the taphole is again plugged and the air blast is turned on.

From this point the operation of the cupola is a semi continuous batch-type process. As metal is melted and removed, additional charges of coke and iron are added through the charging door. It is desirable to maintain the charge up to the level of the charging door at all times to obtain uniform combustion and melting condition. The tapping process may be continuing upto as long as fourteen to sixteen hours.

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